BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Two-Dimensional Arrays and Records

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. For s = 'COMPUTER', what is s.substring(1,3) in OCR ERL?

2. Which is an output from a greeting program?

A) A typed name
B) A mouse click
C) A displayed greeting
D) A key press

3. "Duplicating the same long code block everywhere always improves maintainability." Correct this. Explain why.

4. Nested IF statements are hard to read. "Indentation proves the logic is correct." Correct this. How does indentation help?

Today's objectives

Explain the core ideas of two-dimensional arrays and records using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Two-dimensional arrays

A 2D array stores values in rows and columns and can be accessed with two indexes.

Zero-based indexing

In a zero-indexed grid, the first row and column use index 0.

Records

A record groups related fields about one entity under a single item.

Choice of structure

Use a grid for table-like values and a record for named fields such as name and age.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Database table	Each horizontal row is one pupil record. ID, Name and Mark are fields (columns).
Two-dimensional array	A fixed-size array emulates rows and columns; pupils[row,column] identifies one cell using zero-based indexes.
Data types	An OCR ERL array stores one element type. In a string-based table model, '72' is text; convert it before a numerical comparison.
Records	A named record can group an integer ID, text Name and integer Mark for one pupil without relying only on column positions.

I Do - One worked example

Use the Pupils table: ID 101 Ada 72; ID 102 Ben 43; ID 103 Mia 81. In the string-based 2D array, what is pupils[0,1]? Explain the two indexes.

1. Label rows 0, 1 and 2 for Ada, Ben and Mia. Columns 0, 1 and 2 hold ID, Name and Mark.
2. The FIRST index 0 selects Ada's record (row 0), not column 0.
3. The SECOND index 1 selects the Name field (column 1).
4. pupils[0,1] is 'Ada'. Both indexes start at 0; marks are represented as text in this particular array.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Using the SAME Pupils table, identify the value in pupils[2,1] and explain the indexes. [3 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: The rows represent pupil records, with name in column 0 and attendance status in column 1. Complete the update to mark Ben present, then predict the output.

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "Present"
pupils[1,0] = "Ben"
pupils[1,1] = "Absent"
pupils[1,1] = ...
print(pupils[1,1])
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. In the Pupils table, state the Name stored at row index 1, column index 1. [2 marks]

Q3. Explain what a row (record) and a column (field) represent in the Pupils table. [4 marks]

Q4. Describe one difference between a 2D array of text and a named pupil record. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A 2D array uses two indexes to locate an element. ☐ T ☐ F
2. Each horizontal row is one pupil record ☐ T ☐ F
3. A fixed-size array emulates rows and columns; pupils[row,column] identifies one cell using zero-based in... ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 OMP; indexing starts at zero and the second argument gives length. ☐

Q2 C - A displayed greeting ☐

Q3 Related logic can be reused, understood, tested and changed in one place. ☐

Q4 Indentation shows blocks and nesting. Some languages require it. It helps people read and change code, but does not prove the logic is correct. ☐

Code First - Model

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "Present"
pupils[1,0] = "Ben"
pupils[1,1] = "Absent"
pupils[1,1] = "Present"
print(pupils[1,1])
```

Output: Present

We Do - Check the model

1. pupils[2,1] = 'Mia' [1]; index 2 selects third row [1]; index 1 selects Name column [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Ben [1]; the first index selects row 1 and the second index selects Name/column 1 [1]. ☐

Q3 A row stores the data for one pupil [1] across that pupil's related values [1]. A field/column stores the same property, such as Mark [1], for all pupil records [1]. ☐

Q4 The 2D array is accessed by row and column indexes [1] and uses one consistent element type [1]; a named record uses field names and may store different field types [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: